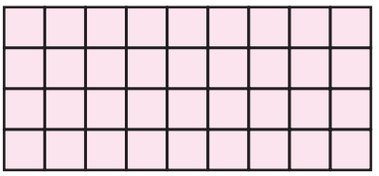
Activity

Form all possible rectangles with area 36 sq.cm by completing the following table.

Length	36	18		9
Breadth	1		3	



Observe and answer the following

(i) When length decreases, the breadth _____

(ii) When breadth increases, the length _____

(iii) If the length is 8 cm what will be the breadth? -Discuss.

Extend this activity and try the same with area 24 and 48 sq.units.

Example 4.5

60 workers can spin a bale of cotton in 7 days. In how many days will 42 workers spin it?

Solution

Let x be the required number of days. The decrease in number of workers lead to the increase in number of days. (Therefore, both are in inverse proportion)

Number of workers	60	42
Number of days	7	x

For inverse proportion $x_1y_1 = x_2y_2$

Hence $60 \times 7 = 42 \times x$

$$42 \times x = 60 \times 7$$

$$x = \frac{60 \times 7}{42}$$

$$x = 10$$

In 10 days 42 workers can spin a bale of cotton.

Example 4.6

The cost of 1 box of tomato is ₹ 200. Vendan had money to buy 13 boxes. If the cost of the box is increased to ₹ 260 then how many boxes will he buy with the same amount?

Solution

The cost of one box = ₹ 200

Increased cost of one box = ₹ 260

Let x be the number of boxes bought by Vendan.

As cost of boxes increases the number of boxes decreases.

This is in inverse proportion. Therefore, $x_1y_1 = x_2y_2$

Number of boxes	13	x
Cost in ₹	200	260

$$13 \times 200 = x \times 260$$

$$x \times 260 = 13 \times 200$$

$$x = \frac{13 \times 200}{260}$$



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